WindBrowser manual

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1 WindBrowser

This manual describes WindBrowser 0.9.80-b.

Suomenkielinen ohje (PDF, Finnish manual): https://pclusius.github.io/windbrowser-docs/WindBrowser-manual-fi.pdf

WindBrowser is an Android app that downloads wind forecasts from the Finnish Meteorological Institute (FMI) and, with your own free key, Météo-France for map areas you choose, and **keeps every forecast run** on the phone. It is meant for judging how far a forecast can be trusted, for example before a sail race or a cruise leg on the Baltic Sea.

1.1 Why keep every run?

FMI serves only the latest run of each model. When a new run replaces the old one you cannot see how the forecast has changed. WindBrowser stores each run it downloads, so you can compare successive runs for the same place and time. If they agree, the forecast is probably reliable; if they differ a lot, it is not.

History exists only from the moment the app starts downloading, so add your area a few days before you need it.

1.2 Models

Model	Resolution	Area size limit	Data per square degree and run
HARMONIE	fine (about 2.5 km)	16 square degrees	about 385 kB
ARPEGE (Météo-France)	medium (0.1 degree, about 11 km)	400 square degrees	about 35 kB
ECMWF-IFS	coarse (0.25 degree)	1600 square degrees	about 2 kB

HARMONIE covers northern Europe. ARPEGE covers Europe and the Baltic and needs your own Météo-France key (ARPEGE). ECMWF-IFS works anywhere in the world. They can be combined: ECMWF-IFS under HARMONIE gives a wide overview with detail where you need it, and a third model shows how far the models agree.

1.3 How to read this manual

The chapters follow the order in which you will use the app:

1. First start: choose your first area.
2. The map: the wind layer and controls.
3. Charts: tap a point and compare runs.
4. Tiles: manage your areas.
5. ARPEGE and the key: Météo-France's model and its key.
6. Stations: observed wind.
7. Settings
8. Updates and troubleshooting

9. About and credits

!!! note The app is for use over the sea. Observation stations are a curated set of marine stations.

2 First start

When you open WindBrowser for the first time the map is empty. **Nothing is downloaded until you have chosen an area.**

The menu (hamburger button, top left) holds **Add area here...** at the top:

2.1 Add an area

1. Open the menu with the **hamburger button** at the top left of the map.
2. Pick **Add area here...**.
3. Choose the **model**: HARMONIE (detailed, small areas), ARPEGE (medium, Europe and the Baltic, needs your own free Météo-France key, see ARPEGE) or ECMWF-IFS (coarse, large areas, very small downloads).
4. A rectangle appears inside the current view. Drag its **top-left and bottom-right handles** to cover the sea area you want. Touches elsewhere pan the map.
5. The dashed outline around your rectangle is the area that will really be downloaded. It is slightly larger so that points at the edge of your rectangle can still be interpolated.
6. Check the **disk estimate** in the panel. A warning appears if the data limit would be exceeded.
7. Confirm. The first forecast starts downloading ("Downloading the first forecast..."). When data is ready, the panel under the map says "Tap the map, or an orange station").
8. The app now asks once whether it may **run without battery restrictions** ("Keep the forecasts coming?"). Allow it if you want no gaps in the forecast history; see Updates. You can change it later in Android's settings: App info → Battery → Unrestricted.

!!! tip For a sail race or just a cruise leg, choose a HARMONIE area around the course and a larger ECMWF-IFS area around it. Both can be active at once.

2.2 Size limits

An area can hold at most 16 square degrees with HARMONIE, 400 with ARPEGE and 1600 with ECMWF-IFS. Larger rectangles are refused. See Tiles for what happens when areas overlap.

2.3 What happens next

The app checks for a new forecast every hour in the background, and also when you bring it to the front (at most every 10 minutes). See Updates.

3 The map

3.1 What you see

- **Wind layer**: colours show wind speed, from blue (calm) through green, yellow, orange and red to purple and black (35 knots and more). Wind barbs show direction and speed in knots. The layer is opaque.
- **Colour bar** with the numbers inside it.
- **Temperature map**: choose **Temp** on the toggle in the speed chart and the colours show the 2 m air temperature from -20 °C (black) through purple, blue, cyan, green, yellow and orange to red and light grey at 40 °C and above (the `nipy_spectral` scale), with its own colour bar in °C. The wind barbs stay. ARPEGE and ECMWF-IFS have temperature, HARMONIE does not: the colours come from ARPEGE where it covers the point and otherwise from ECMWF-IFS, and the barbs from the topmost model, so where a HARMONIE tile lies over one of them the colours come from that one and the barbs from HARMONIE; a HARMONIE tile with nothing under it shows barbs only. The grey land setting is kept as it is, so with grey land on, the temperature shows over the sea only. Choose **Speed** to go back to the wind layer.

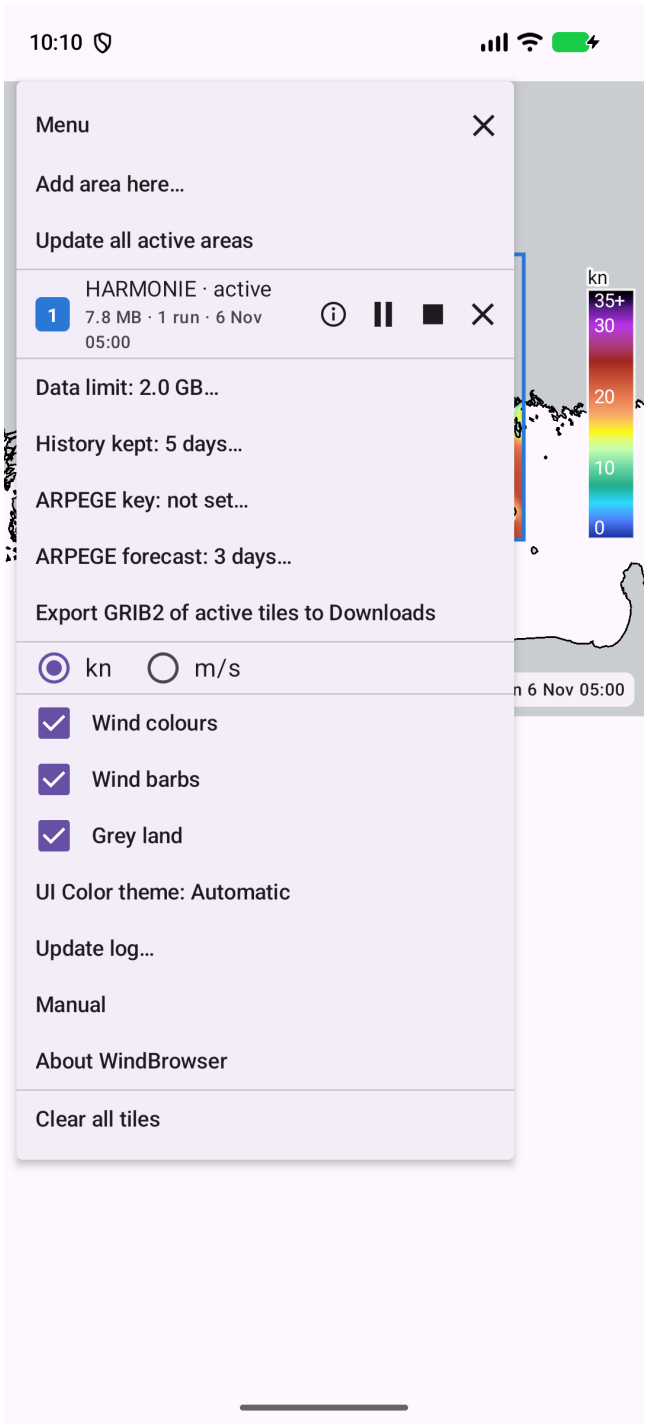


Figure 1: Menu

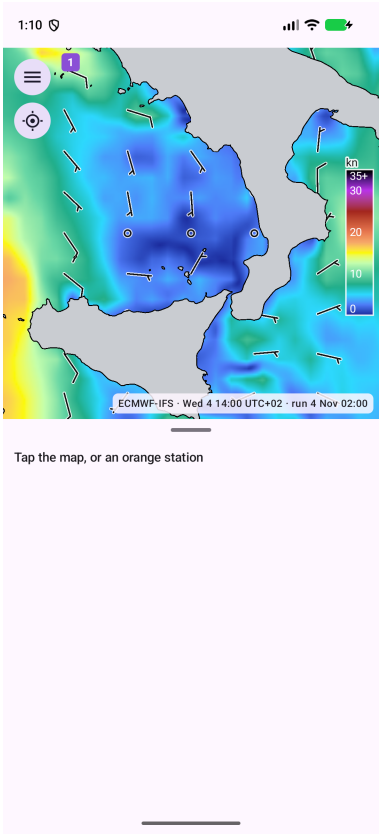
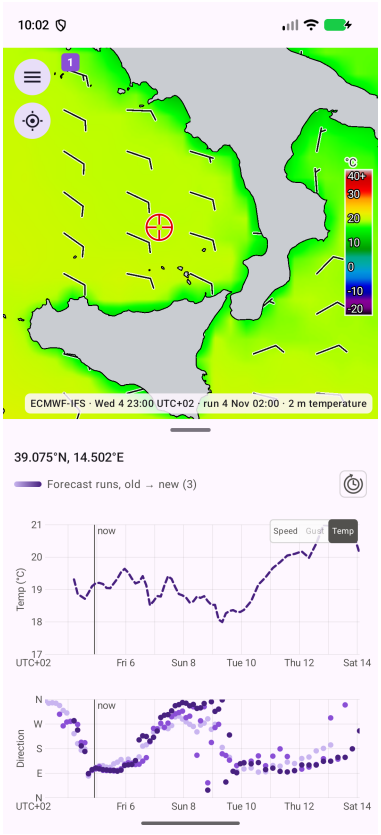


Figure 2: Map overview



- **Coastline**: grey land with a black shoreline, drawn from a bundled vector map, so no map tiles are needed and the map works offline. - **Place names**: small names of larger towns; more appear as you zoom in. - **Station dots**: orange dots are observation stations (Stations); small when zoomed out, larger when zoomed in. - **Tile outlines**: the borders of your areas, with a number badge. Solid means active, dashed means paused or stopped. Blue is HARMONIE, green is ARPEGE, purple is ECMWF-IFS.

Zoom runs from level 5 to 15. Drag to pan, pinch to zoom.

3.2 The caption

The caption at the bottom of the map reads, for example, `ECMWF-IFS · Sat 7 20:00 UTC+02 · run 4 Nov 02:00 · tap for now`: the model, the forecast time shown and the model run it comes from. In the temperature map it also says "2 m temperature" and, where the barbs come from another model, "wind barbs FMI-HARMONIE". For a tapped point it names the model that covers the point, and adds "also ECMWF-IFS" when a second model reaches it too. When a point is selected, tap the caption (**tap for now**) to put the chart cursor at the actual time of day; the map shows the nearest forecast hour.

3.3 Old data fades

A tile whose newest run is more than 24 hours old fades to grey, fully grey at 48 hours. Grey means the data is stale, usually because downloads are paused or failing.

3.4 Two models together

The finest model is drawn on top: ECMWF-IFS below ARPEGE below HARMONIE, so you see the fine model where it exists and the coarser ones around it. To look at a layer that is covered, hide the ones above it by tapping their names in the chart legend (Charts).

3.5 Location button

The button under the menu button finds your position (one fix; GPS if you allowed precise location), selects that point as if you had tapped it, and centres the map there. The position is not stored or sent anywhere.

3.6 Divider

Between map and charts there is a drag handle. Portrait: map above, charts below. Landscape: map on the left, charts on the right. Each orientation remembers its own split.

4 Charts

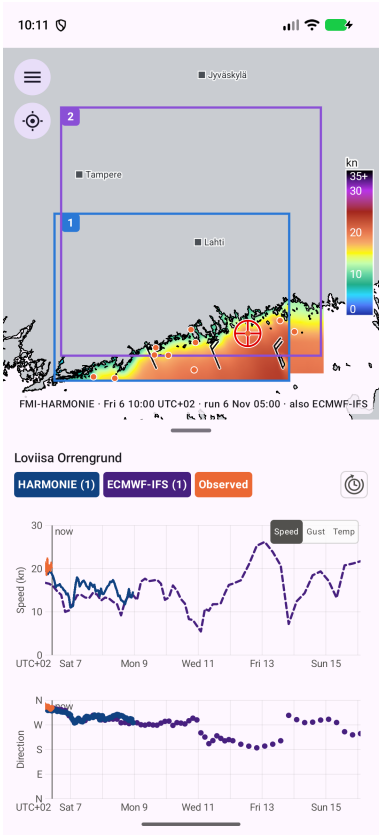


Figure 3: Charts

Tap anywhere on the map to select a point (a red crosshair marks it), or tap an orange station dot; the title then shows the station name or the coordinates. The charts show the forecast at that point from **every stored run** of every area that covers it, whether the tile is active or not.

4.1 Reading the charts

- **Speed chart** (top): each run is one line. HARMONIE is blue and solid; ARPEGE is green with long and short dashes; ECMWF-IFS is purple and dashed. Older runs are paler, the newest is darkest. With only one model the legend reads "Forecast runs, old → new".
- **Direction chart**: shown as dots, never lines, because 359 and 1 degrees are neighbours.
- **Speed / Gust / Temp toggle**: small switch in the corner of the speed chart. Gust exists for HARMONIE and ARPEGE and temperature (2 m air temperature, in °C) for ECMWF-IFS and ARPEGE; each option is greyed out where no run has it. Choosing **Temp** also turns the map's colours to temperature (see The map). If you tap a point that no temperature forecast covers, the chart and the map go back to speed.
- **Observed wind** from a tapped station is drawn in orange. With **Temp** chosen it is the observed air temperature (FMI, SMHI, DMI and Marine Helsinki stations).

If the runs lie close together the forecast is consistent. A wide fan means uncertainty.

4.2 The cursor and the readout

Touch a chart with **one finger** to move the cursor. The map follows the cursor in time. The line above each chart reads, for example:

`HARMONIE: 12.2 (11.1–13.0)  ARPEGE: 13.4 (12.0–14.1)  ECMWF-IFS: 15.2 (11.1–15.2)  Observed: 13.2`

Before you touch the chart the models show a dash (–).

For each model it shows the value of the newest run covering that time, and in brackets the lowest and highest of all runs covering it, then the observation.

4.3 Legend

The legend shows each model's name in bold and in the colour of its lines, with its number of runs, and "Observed" in orange. **Tap a name** to hide that model: its lines, dots and readout leave the charts, and its **map layer, wind barbs and caption leave the map**, which is the way to see a layer that lies under another one. Tap again to show it. The names turn grey while a model is hidden. Choosing another point shows everything again.

4.4 Zoom

Two fingers zoom and pan the shared time axis. Speed rescales to what is in view and hourly ticks appear when zoomed in. **Tap for now** resets the axis.

4.5 Auto-advance

The clock button with the circular arrow at the right end of the legend moves the cursor, and so the map, forward by itself: a quarter of an hour per quarter second, from the cursor (or from now) to the end of the axis in view. A touch on a chart, the button again, or "now" stops it.

4.6 Sun and moon

The place in the title above the charts, the coordinates or a station's name, is underlined: **tap it** to see the sun and moon for that place.

- The pop-up opens on the day of the chart cursor, or on today. The left and right arrows step one day at a time, and **Today** returns to today.
- The **day strip** runs from midnight to midnight and is coloured by how far the sun is below the horizon: night, astronomical, nautical and civil twilight, and day. Red marks show sunrise and sunset, the dot is solar noon, and the bar under the strip shows when the moon is up, with its rise and set marked in red.
- The **table** gives the length of daylight (sunrise to sunset) and of the civil, nautical and astronomical day: the time the sun is more than 6, 12 or 18 degrees below the horizon is the twilight, so the dark time of each kind is the rest of the 24 hours. Where the sun does not cross a level, as on a Baltic summer night, the table says "all day" or "never".
- The **moon** is drawn as it looks that day (flipped in the southern hemisphere), with the phase, the share lit, its age in days, moonrise and moonset, and the next new and full moon.

Everything is calculated from the position and the date, so it works without any forecast or network. The sun is accurate to about 2 minutes and the moon to about 5. The times are on the phone's clock and its UTC offset, and a "day" is the phone's calendar day.

4.7 Units

Speed is shown in knots or m/s (Settings). Times show the phone's UTC offset, for example `UTC+02`.

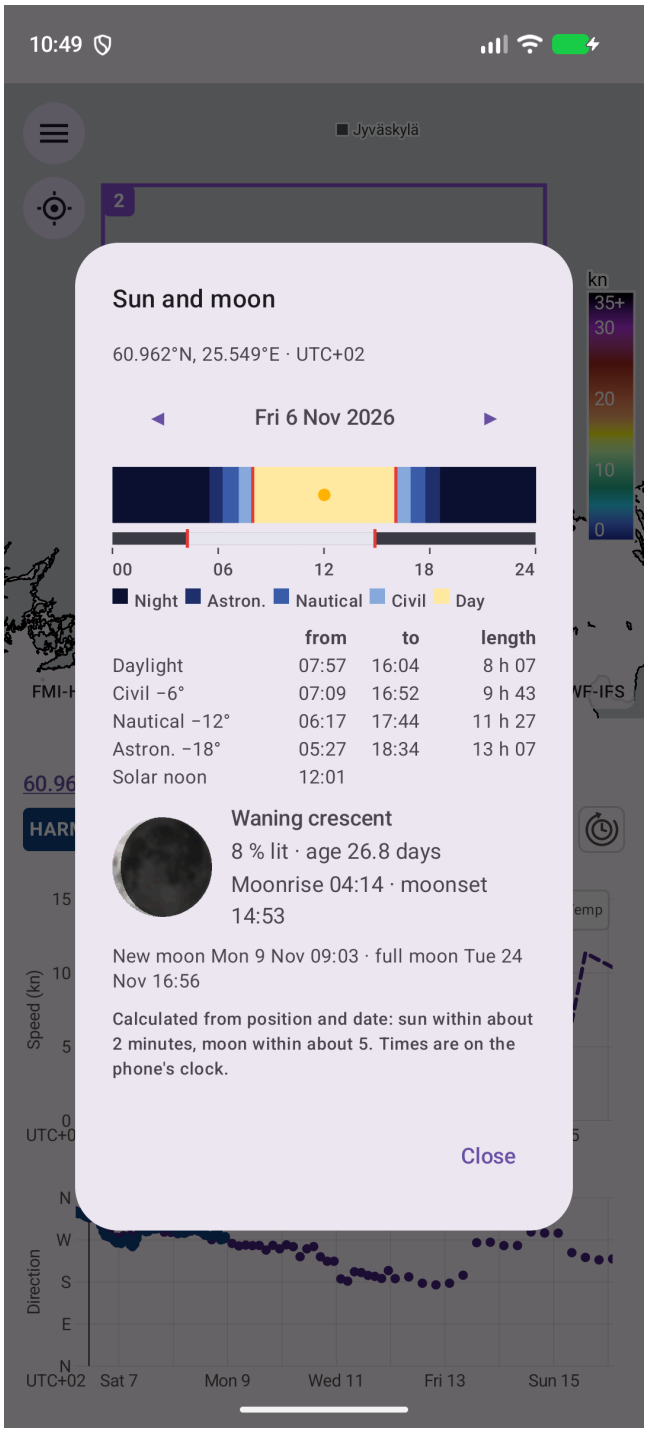


Figure 4: Sun and moon

5 Tiles

An area you download is called a **tile**. You can have as many as you like, numbered from 1 (the data limit is what bounds them). Everything is managed from the hamburger menu.

5.1 The tile list

Each row shows the tile's number (blue badge HARMONIE, green ARPEGE, purple ECMWF-IFS), model, state, size, number of runs and newest run, for example `HARMONIE · active · 4.9 MB · 9 runs · 4 Nov 08:00`. Tap a row to zoom the map to that tile.

Button	Action
i	List the stored runs (model run time and file size)
pause / play	Pause stops downloads but keeps the data; play resumes
stop	Stops downloads; the tile is removed when its data has expired
delete	Removes the tile and its data. Always asks for confirmation

States: **Active** downloads; **Paused** keeps data but does not download; **Stopped** is removed once its data has expired.

5.2 Overlap

Only active tiles of the same model count.

- An area completely covered by existing tiles is refused.
- If 70 % or more is covered by one tile, the app offers to **enlarge that tile** to hold both, or to add the area anyway.
- A smaller overlap is accepted with a note that the overlap is downloaded twice.

5.3 Storage limits

- **Data limit**: 1 to 5 GB (default 2). When exceeded, the oldest runs go first, those of non-active tiles before active ones; the newest run of an active tile is never removed. A dialog tells you once what was removed and what you can do: shorter history, smaller areas or fewer tiles, ECMWF-IFS, or a higher limit.
- **History**: 1 to 14 days.

Clear all tiles always asks for confirmation.

6 ARPEGE and the Météo-France key

ARPEGE is the global forecast model of Météo-France. WindBrowser uses its 0.1 degree (about 11 km) version over Europe and the Baltic, with wind, **gusts** and **2 m air temperature**. It gives four runs a day (00, 06, 12 and 18 UTC) and forecasts up to 102 hours ahead.

Unlike FMI's models, Météo-France gives its data only to users who have **their own free key**. The key is kept on your phone only and is sent only to Météo-France. Without one, ARPEGE cannot be chosen when you add an area, and an ARPEGE tile that is already there says it needs a key.

6.1 Get a key

1. Make a free account at **portail-api.meteofrance.fr**.
2. Subscribe to the API called **API Modèle ARPEGE**.

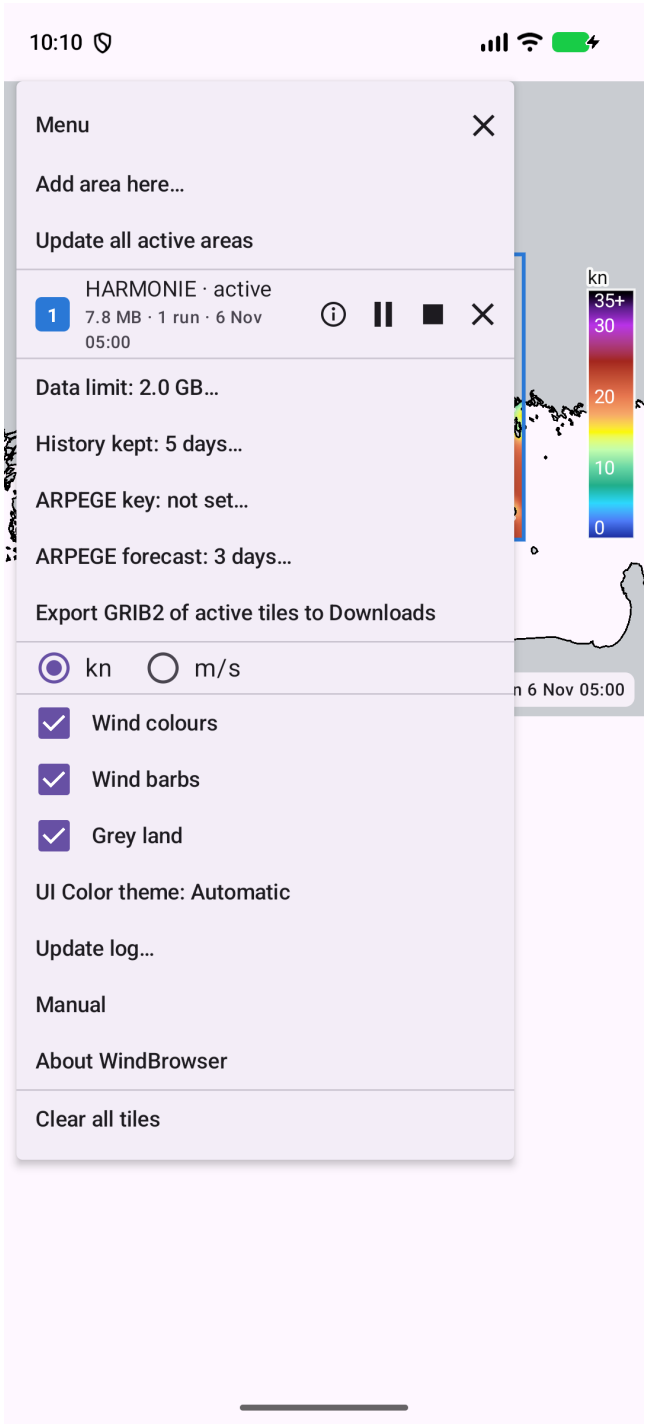


Figure 5: Tile list

3. On the API's page, under *Configure API*, choose the token type **API Key**, set how long it should last (365 days works) and press **Generate Token**.
4. **Copy all of the key.** It is one very long line of text, about 1,500 characters, with no spaces. Click in the key field, use *select all* and copy; a double click selects only a part of it.

!!! note Making a new key on the portal may stop the old one from working. If downloads stop with "The Météo-France API key was rejected", the key has expired or been replaced: make a new one and enter it again.

6.2 Enter the key in the app

Menu → **ARPEGE key: not set…**, paste the key and press **Save**. The dialog shows how many characters it got and refuses a key that is clearly too short, which usually means it was cut when copied. After saving, the menu says **ARPEGE key: set…**; open it again to replace the key, or press **Remove** to delete it. If an ARPEGE tile is waiting, an update starts at once.

6.3 Add an ARPEGE area

Choose **ARPEGE** (green) when you add an area (First start). An area can hold at most 400 square degrees, and the data is small, about 35 kB per square degree and run for three days of forecast. ARPEGE covers 20°N to 72°N and 32°W to 42°E.

6.4 How far ahead

Menu → **ARPEGE forecast: 3 days…** sets how far ahead ARPEGE is downloaded: 1 to 4 days, 3 by default. Météo-France sends one small file for every variable and hour, so a longer forecast means more requests (3 days is about 290) and a longer update. The portal allows 100 requests a minute and the app spaces them out, so the first download of a run takes about **three minutes**. A change applies from the next run; what is already stored stays.

6.5 What to expect

ARPEGE is drawn with green lines in the charts, with long and short dashes, and its tile outline and map layer are green. It is coarser than HARMONIE and finer than ECMWF-IFS, so on the map it lies between them. Gust and temperature work for ARPEGE as for the other models (Charts).

7 Stations

The small orange dots on the map are marine observation stations. Tap one to see its **measured wind** (and, where the station has a thermometer, its **air temperature** on the Temp chart) drawn on the charts together with the forecasts, so you can see how well recent runs matched reality.

7.1 Sources

Source	Data
FMI	10-minute observations
Marine Helsinki	Kruunuvuorenselkä
SMHI (Sweden)	hourly
DMI (Denmark)	10-minute; gust is the highest 3-second mean
TalTech	Tallinnamadal, latest values only
Estonian Environment Agency	12 coastal stations: 10-minute history of the last few days (the agency keeps about 3.7 days); the other stations: current wind only, as a text label

Tallinnamadal and the Estonian stations that are not among the 12 coastal ones show only the wind of the moment as a label on the map, not a chart, because those sources offer no history. The chart never reaches further

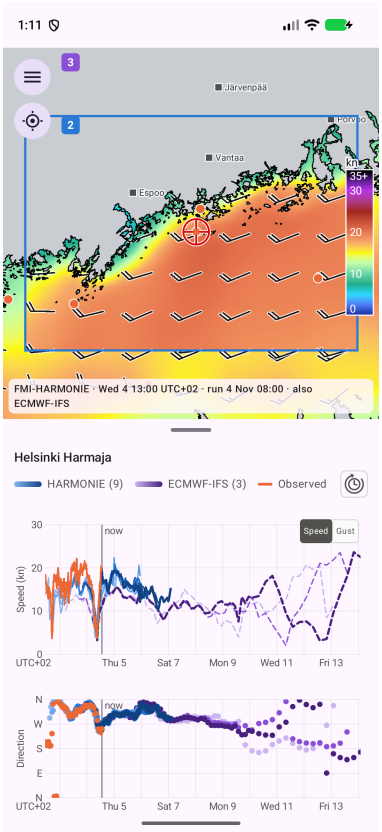


Figure 6: Station

back than the oldest forecast run at the tapped point. Where a source has no gust or direction, those lines are left out. A gap in the line means the station was silent.

Station dots are 2 dp in a wide view and grow to 12 dp from zoom level 10.

8 Settings

All settings are in the hamburger menu.

Setting	Meaning
Update all active areas	Starts an update now
Data limit	1–5 GB (default 2)
History kept	1–14 days of runs to keep (applied immediately)
ARPEGE key	Your Météo-France key, needed for ARPEGE; see ARPEGE
ARPEGE forecast	How many days ahead ARPEGE is downloaded, 1–4 (default 3), from the next run on
Export GRIB2 of active tiles to Downloads	Copies the newest raw GRIB2 of every active tile to Downloads, with the tile number in the name (Android 10 and newer)
kn / m/s	Speed unit, default knots. Barbs are always in knots
Wind colours, **Wind barbs**	Tick each layer on or off
Grey land	Untick to show only the shoreline
UI Color theme	Automatic, light or dark; tap to cycle
Update log…	See Updates
Manual	This manual, stored in the app, so it works without a connection
About WindBrowser	Credits, version, and links to the online manual and the PDF
Clear all tiles	Removes everything; always asks first

9 Updates and troubleshooting

9.1 How updates work

- A background job runs about **hourly** and updates every active tile.
- The app also updates when you bring it to the front, at most every 10 minutes.
- Because Android may put the hourly job to sleep while the phone idles, an **alarm every hour** also asks for an update.
- FMI keeps **only its latest run**, and the app takes only the newest run of ARPEGE too. A run missed because the phone was idle cannot be fetched later, so open the app now and then in the days before you need it.
- A new ARPEGE run takes about three minutes to download, because Météo-France sends it in many small requests (ARPEGE).

9.2 Battery use: set the app to Unrestricted

Android saves power by delaying background work, most of all overnight, and a forecast run that appears meanwhile is lost for good, because FMI keeps only the latest one. Letting WindBrowser run without battery restrictions fixed this in practice: the updates arrive on time and the forecast history has no gaps.

The app asks about this **once**, right after you add your first area. **Allow** shows Android's own question ("Let app always run in background?"). The app checks for new forecasts only about once an hour and downloads only when there is a new run, so the extra battery use is small. If you chose **Not now**, or want to undo it, change it

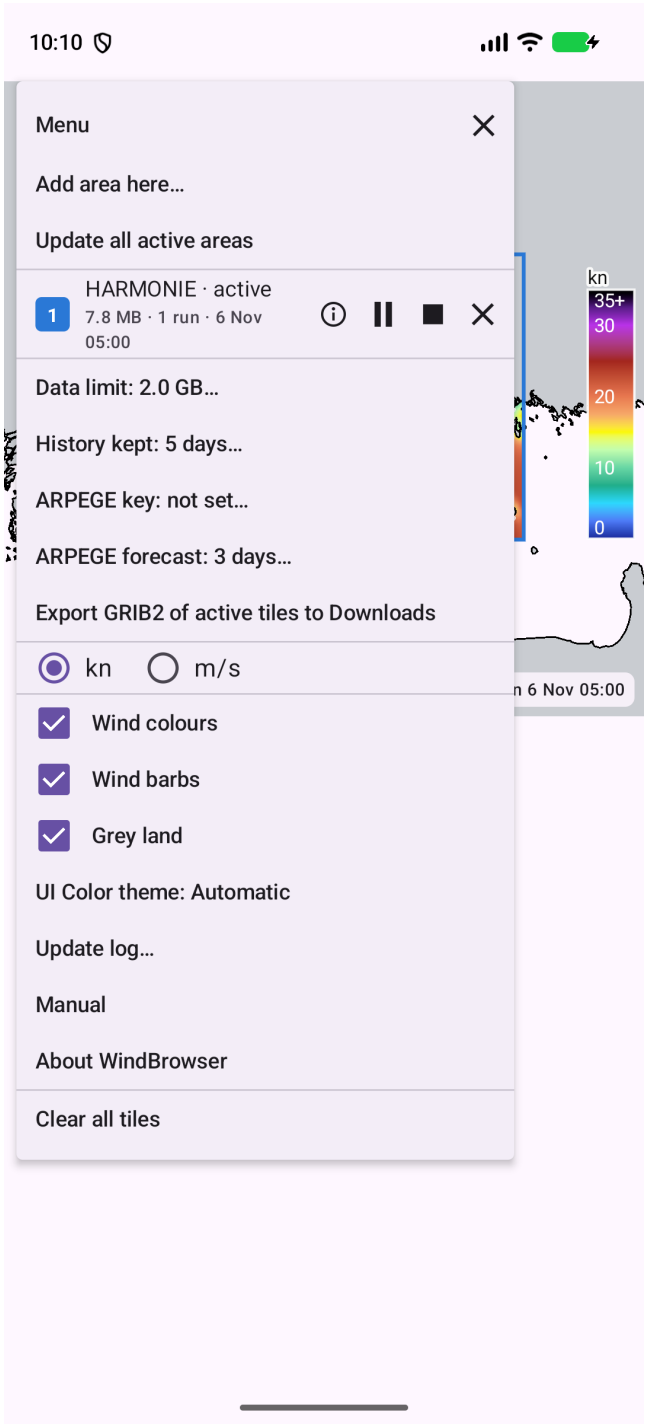


Figure 7: Menu

any time in Android's settings: **App info → Battery → Unrestricted** (the wording differs between phone makers). The update log tells whether the app is unrestricted at the moment.

9.3 Update log

Menu → **Update log…** lists each job (hourly, alarm or manual, attempt, standby bucket, battery unrestricted or optimised, dozing or not), the result for each tile, and the end. Use it to find out why a run is missing.

9.4 Troubleshooting

Problem	What to try
"Downloading the first forecast…" does not finish	Check the connection; see the update log
Tile is grey	Its newest run is older than 24 h. Check that it is active and that the phone is online
"No forecast at the tapped point"	The point is outside every tile. Add an area (ECMWF-IFS works anywhere)
Runs disappeared	The data limit or history limit pruned them; raise it
HARMONIE area refused	Larger than 16 square degrees, or fully covered by existing tiles
An ARPEGE tile says it needs a key, or "key was rejected"	Enter a key, or a new one if it expired or was replaced: ARPEGE
Android delays updates, or runs are missing overnight	Set the app's battery use to Unrestricted (see above); opening the app also triggers an update

10 About and credits

WindBrowser keeps forecast runs for comparison. The version is shown in the menu under **About WindBrowser**.

10.1 Where to find the manual

- **In the app, offline:** menu → **Manual**.
- **Online (latest):** https://pclusius.github.io/windbrowser-docs/
- **PDF:** https://pclusius.github.io/windbrowser-docs/WindBrowser-manual.pdf, also linked from **About WindBrowser**.
- **PDF in Finnish (suomeksi):** https://pclusius.github.io/windbrowser-docs/WindBrowser-manual-fi.pdf. It is updated only after major changes, so the English manual is the most current.

10.2 Data and credits

- Forecasts: **Finnish Meteorological Institute** open data (HARMONIE and ECMWF IFS), and **Météo-France** (ARPEGE; Open Licence 2.0, through the Météo-France API with your own key).
- Observations: FMI, SMHI, DMI, Marine Helsinki, TalTech and the Estonian Environment Agency (ilmateenistus.ee). Licences are listed in the app.
- Coastline: OpenStreetMap contributors (land polygons) and Natural Earth.
- Place names: GeoNames (CC BY 4.0).

No account is needed. Your position, if you use the location button, is never stored or sent.